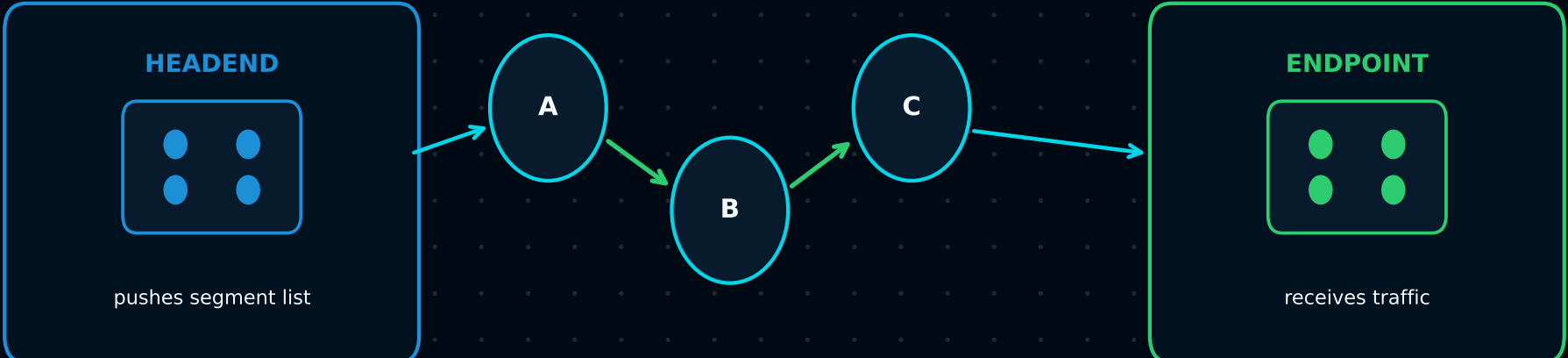


SEGMENT ROUTING

SR-MPLS + SRv6: source-routed transport without RSVP-TE complexity



WHY SR MATTERS

- ✓ **Headend state**
core avoids per-tunnel state
- ✓ **Traffic engineering**
explicit paths without RSVP-TE
- ✓ **Fast reroute**
TI-LFA protects locally
- ✓ **Service steering**
policies map traffic to intent

KEY TECHNOLOGIES

- 1 **SR-MPLS**
label stack
- 2 **SRv6**
IPv6 SID + SRH
- 3 **SR Policy**
color + endpoint
- 4 **Binding SID**
policy as one SID
- 5 **Flex-Algo**
constraint-based SPF
- 6 **PCE**
central path compute

MAJOR USE CASES

- ✓ **DCI / WAN TE**
choose latency or cost paths
- ✓ **5G transport**
slices and deterministic paths
- ✓ **Cloud edge**
steer apps to service nodes
- ✓ **Fast restoration**
sub-50ms local repair

OPERATIONS CHECKS

- ✓ IGP advertises SIDs
- ✓ SRGB consistent across nodes
- ✓ BSID unique and installed
- ✓ BGP color / VRF steering works
- ✓ Counters increment on policy
- ✓ ping/traceroute validates path

DESIGN WATCHOUTS

- ! SRGB mismatch
- ! label/SRH depth limits
- ! Adj-SID instability
- ! MTU and ASIC support
- ! policy up but no steering

SR is the transport toolkit for traffic engineering, fast reroute, and service steering.

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